

Temporal graphs: parameters, labelings and realizations

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Plan

- 1 Introduction
- 2 Parameterized Restless Temporal Path
- 3 Labeling schemes
- 4 Temporal graph realization

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Temporal graphs: motivations, history

Motivations:

- Graphs represent relationships between entities
- In many real-world applications, those relationships are time-related:
 - Public Transit network: there is a bus that goes directly from stop s to stop t only at time $\tau_1, \tau_2, \tau_3 \dots$
 - Phone call network: Alice calls Bob from time τ to time τ'
 - Social network: Alice follows Bob at time τ
 - ...

Temporal graphs: motivations, history

Motivations:

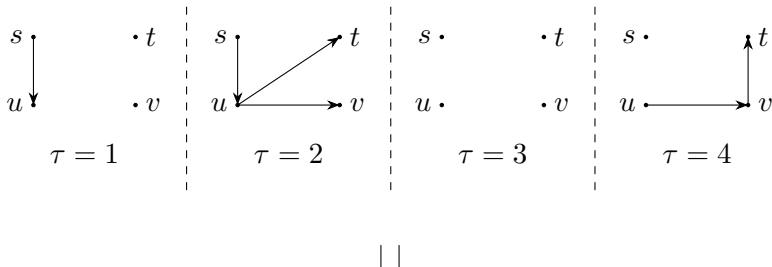
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History:

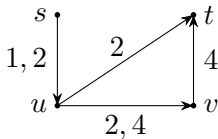
- Fragmented literature, plenty of models
- Time-dependent network (e.g.: road networks [Dreyfus 1969])
- Evolving graph (e.g.: ad-hoc networks [Xuan et al. 2003])
- Temporal network (e.g.: social network [Holme et al. 2011]), link stream [Latapy et al. 2017]
- Time-varying graph (ex: distributed computing [Casteigts et al. 2012]), temporal graph (ex: graph theory [Michail 2015])

Temporal graphs: models

Sequence of snapshots:

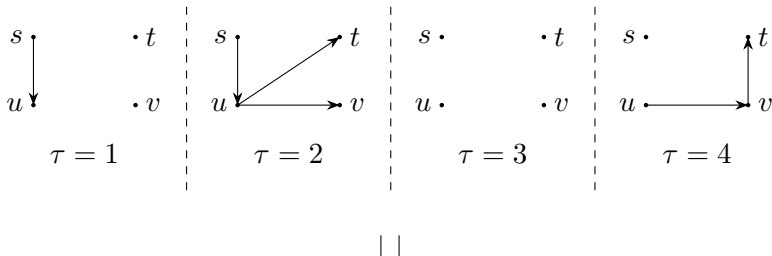


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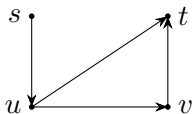


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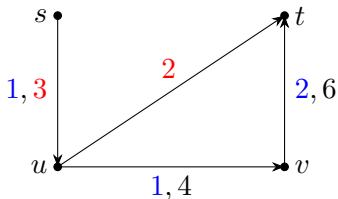
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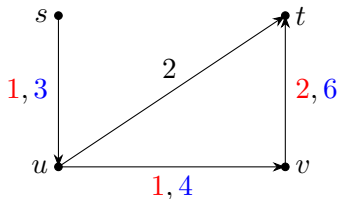
Underlying graph

Temporal paths: non strict and strict settings

Non-strict/Strict temporal path: A path and an **non-decreasing/increasing** sequence of times at which arcs can be traversed one after the other



Non-strict setting



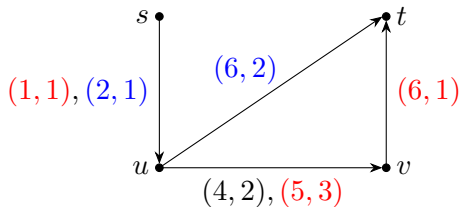
Strict setting

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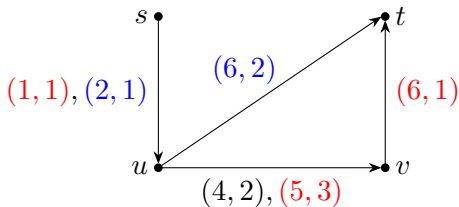
The model

- Directed graph
- Delay: time it takes to traverse an arc



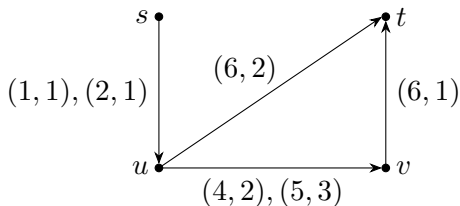
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- Directed graph
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- Uniform delay zero = non-strict setting
- Uniform delay one = strict setting

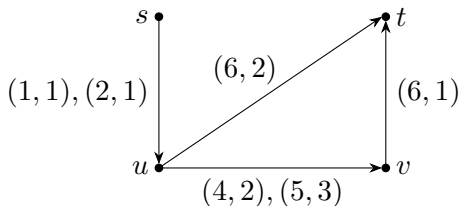
The model



$$[(s, u, 1, 1), (s, u, 2, 1), (u, v, 4, 2) \dots]$$

- Each time we can traverse an arc $\Rightarrow$ a timed arc
- **Timed arc**: an arc, an appearance time, a delay (e.g. $(u, v, 4, 2)$ is a timed arc)

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- Each time we can traverse an arc $\Rightarrow$ a timed arc
- **Timed arc**: an arc, an appearance time, a delay (e.g. $(u, v, 4, 2)$ is a timed arc)
- Input: sorted list of timed arcs in non-decreasing order of appearance time
- M : number of timed arcs $\Rightarrow$ size of the **input** in $\mathcal{O}(M)$

Generalization of static graph problems

- Revisit classical static graph problems in the context of temporal graphs
- Usually **more difficult**

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Definition

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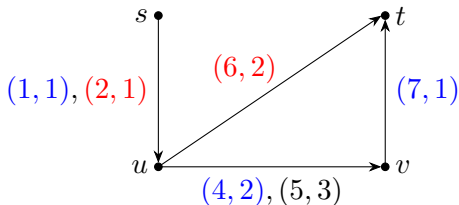
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- With static graphs: find SCC in linear time
- With temporal graphs: find TSCC **NP-hard**

The problem: Restless Temporal Path

Δ -restless temporal path: Cannot wait more than Δ at an intermediate node



$$\Delta = 2$$

The problem: Restless Temporal Path

RESTLESS-TEMPORAL-PATH

Input: A temporal graph G , a source node s , a target node t and a waiting time $\Delta \in \mathbb{N}$

Question: Is there a Δ -restless temporal path from s to t in G ?

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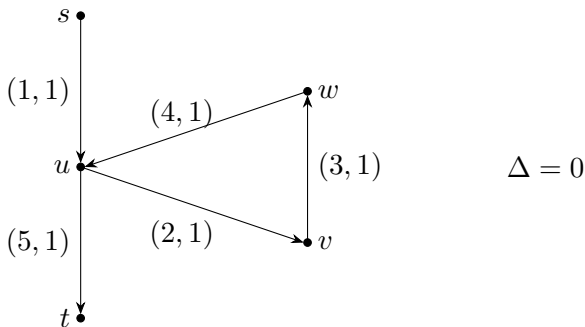
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Applications:

- Detection of a chain of infections in the SIR (Susceptible-Infected-Recovered) model
- Public transit networks

The problem: Restless Temporal Path

- Computing a restless temporal **walk**: **polynomial time** [Bentert et al. 2020]



- Computing a restless temporal **path**: **NP-hard** [Casteigts et al. 2021]

Parameters

NP-hard problems $\Rightarrow$ find parameters that **bring tractability**

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Restless temporal path:

- Output-related parameter: the number of hops of the temporal path [Casteigts et al. 2021, Zschoche 2023]
- Classical parameter applied to the underlying graph: the feedback edge number of the underlying graph [Casteigts et al. 2021]
- New 'temporal' parameter: the timed feedback vertex number [Casteigts et al. 2021]

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Invariant by permutation of the snapshots $\Rightarrow$ does not seem to really grasp the **temporal aspect**

The (vertex)-interval-membership-width parameter

Interval-membership-width parameters: [Bumpus and Meeks 2022]

- Novel parameters that capture some temporal aspects
- Promising for reachability problems
 - FPT for temporal Euler circuit
 - NP-hard for MinReachDelete but FPT with the vertex variant
 - Counting the number of temporal paths [Enright et al. 2023]
 - Temporal Firefighter Problem [Hand et al. 2022]

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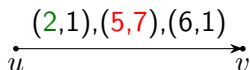
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 - NP-hard for MinReachDelete but FPT with the vertex variant
 - Counting the number of temporal paths [Enright et al. 2023]
 - Temporal Firefighter Problem [Hand et al. 2022]
- Incomparable with the other FPT parameters for the restless temporal path problem

Goal: Study this new parameter for the restless temporal path problem

The (vertex)-interval-membership-width parameter

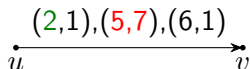
Active window of an arc:



[first departure time of (u, v) , last arrival time of (u, v)] = [2, 12]

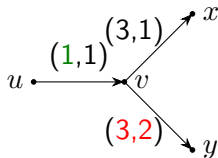
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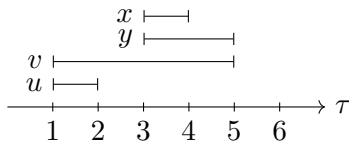


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Active window of a vertex: Smallest window containing all active windows of arcs adjacent to the vertex

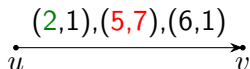


$[1, 5]$



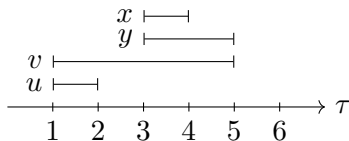
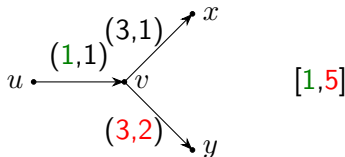
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(vertex)-interval-membership-width: Maximum number of arcs (vertices) active at the same time

Uniform delay one (strict setting)

Proposition

RESTLESS-TEMPORAL-PATH is NP-hard, even if the input temporal graph has interval-membership-width equal to three and uniform delay one.

Uniform delay one (strict setting)

Theorem

When the input temporal graph $G = (V, E)$ has uniform delay one, RESTLESS-TEMPORAL-PATH can be solved in $\mathcal{O}(k2^k M)$ deterministic time, where k is the **vertex-interval-membership-width** of G .

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General idea:

- Scan timed arcs in order of appearance time
- Maintain for each node u the Δ -restless temporal su -paths

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- Solution: store only traces of paths (only the nodes on the path that are active at considered time): at most **2^k traces**
- For each trace, store the **last arrival time** of a Δ -restless temporal path

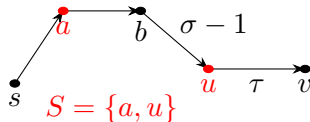
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Details: Scanning timed arcs at time τ :

- Basic step: Extending a found Δ -restless temporal path with timed arc (u, v, τ) :
 - Look at the couples (S, σ) in $L[u]$: $S \subseteq V$ is a trace, σ the last arrival time of an associated path
 - Check:
 - $v \notin S \Rightarrow$ valid temporal path
 - $\tau - \sigma \leq \Delta \Rightarrow \Delta$ -restless
 - Add $(S \cup \{v\}, \tau + 1)$ to $L'[v]$
- Clean up step:
 - Clean up lists by updating the traces and keeping the last arrival time for each trace



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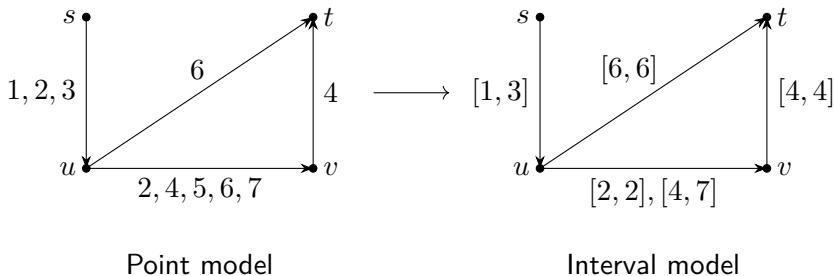
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RESTLESS-TEMPORAL-PATH can be solved in $\mathcal{O}(k2^k M(k + \log M))$ deterministic time, where k is the **vertex-interval-membership-width** of G .

Temporal graphs: interval model



Interval model: More compact $\Rightarrow$ more difficult

Interval model: NP-hardness

Theorem

In the interval model, RESTLESS-TEMPORAL-PATH is **NP-hard**, even if the input temporal graph has **vertex-interval-membership-width** equal to three.

- Reduction from SUBSET-SUM
- Uses intervals of exponential length

Summary and open questions

Summary:

- **FPT algorithms** for the restless temporal path problem with the **vertex-interval-membership-width** in:
 - the point model with uniform delay one
 - the point model with arbitrary (positive) delays (additional computational cost)
- **NP-hardness** in:
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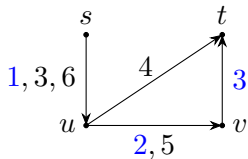
- Interval model with uniform delay one
- What other temporal parameters might bring tractability to the problem and are small in practice?

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Temporal paths: distances

Temporal path: A path and an increasing sequence of times at which arcs can be traversed one after the other

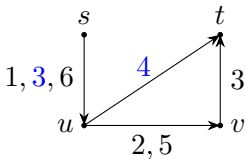


Several notions of distance:

- **Foremost** temporal path: earliest arrival time
- Fastest temporal path: minimum duration
- Shortest temporal path: minimum number of arcs

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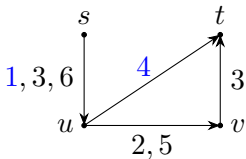


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Labeling scheme for temporal distances

Labeling scheme:

- $f : V(G) \times V(G) \rightarrow \mathbb{N}$ (e.g. connectivity, distance...)
- Encoder: Assigns to each node v a label $L(v)$
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A shortest temporal query from s to t asks what is the minimum number of arcs of a temporal st -path.

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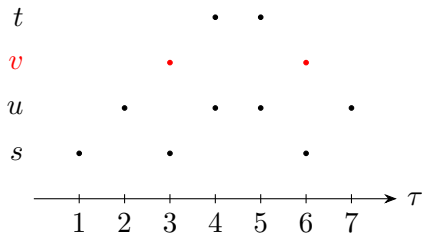
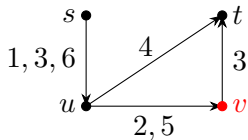
A shortest temporal query from s to t asks what is the minimum number of arcs of a temporal st -path.

State of the art: Labeling schemes only for foremost queries (classical labeling of the time-expanded graph) [Delling et al. 2015]

Goal: Design labelings for shortest, fastest and foremost queries

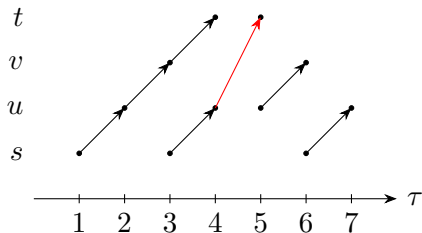
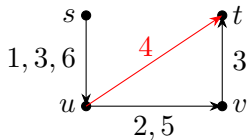
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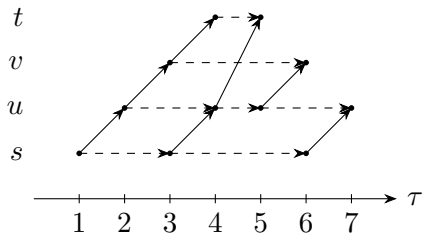
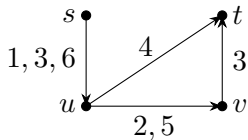
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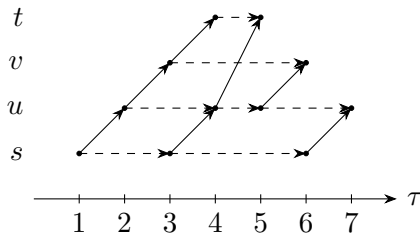
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Labeling scheme for foremost queries

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General idea: [Delling et al. 2015]

- Labeling scheme which answers reachability queries on the time-expanded graph
- (s, t, τ) -earliest arrival time query answering:
 - Find the earliest event (s, τ') that is after τ
 - Find the earliest event (t, τ'') that is reachable from (s, τ') by performing binary search

Contributions and future work

- Designed labeling schemes for the three variants (adaptation of Hub Labeling [Cohen et al. 2003])
- The three variants are **incomparable** in terms of total labeling size
- $\mathcal{O}(\log n)$ -**approximation** in polynomial time

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Future work: Fast **heuristics** for constructing labelings

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Motivations and state of the art

Realization problems

Given a property P , is there a graph that satisfies the property?

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PERIODIC FASTEST TEMPORAL GRAPH REALIZATION

Input: An $n \times n$ matrix D and a period Δ

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- Railway networks, satellite networks (and social networks)
- **NP-hard** [Klobas et al. 2024]
- FPT by the feedback edge number of the underlying graph [Klobas et al. 2024]
- Generalized to up to ℓ labels per period [Erlebach et al. 2024]

Questions

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Model:

- Undirected graph
- No delay

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- Valid for strict and non-strict settings, periodic graphs
- Valid with a **prescribed graph** (if non-strict setting $\Rightarrow \mathcal{O}(n^2 m)$ with m the number of edges of the prescribed graph)

FOREMOST TGR: limitations of the tractability

Theorem

FOREMOST TGR is NP-hard when allowing at most one label per edge.

FOREMOST TGR: limitations of the tractability

Theorem

FOREMOST TGR is **NP-hard** when allowing **at most one label per edge**.

RANGED-FOREMOST TGR

Input: An $n \times n$ matrix D where each entry represents a range $[\ell, r]$

Question: Is there a temporal graph G such that the earliest arrival times match the matrix?

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Theorem

RANGED-FOREMOST TGR can be solved in $\mathcal{O}(k^2 3^k n^4)$ time and $\mathcal{O}(2^k + n^2)$ space, where k is the **number of undetermined entries of D** .

FASTEST AND SHORTEST TGR

Theorem

FASTEST TGR and SHORTEST TGR are NP-hard.

FASTEST AND SHORTEST TGR

Theorem

FASTEST TGR and SHORTEST TGR are NP-hard.

Theorem

FASTEST TGR is W[1]-hard when parameterized by the vertex cover number of the underlying graph plus the largest entry of the matrix.

Future work

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- Does SHORTEST TGR become polynomial-time solvable if we restrict the respective labeling?
- Are there structural parameters for which we can solve SHORTEST TGR in FPT-time? (e.g. treewidth)
- Are there constant factor approximations for FASTEST TGR and SHORTEST TGR? (Under the measurement of fulfilling as many entries as possible)

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Thank you for listening :)

Plan

- 5 References
- 6 Parameterized restless temporal path
- 7 Labeling schemes

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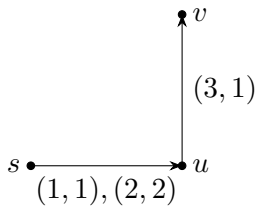
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Plan

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Arbitrary delays



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Hub Labeling (HL)

Assumptions and notations:

- Undirected graph $G = (V, E, c)$ with positive edge costs $c : E \rightarrow \mathbb{R}^+$
- Unique shortest path: $\pi(s, t)$, cost: $c(\pi(s, t))$

Goal: Preprocessing-based techniques

- Faster query answering than Dijkstra
- Subquadratic space consumption

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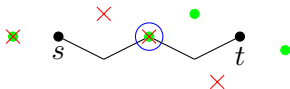
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Hub Labeling (HL): [Cohen et al. 2003]

- Labeling: $L : V \rightarrow 2^V$
- Hub cover property: $\forall s, t \in V, \exists v \in L(s) \cap L(t), v \in \pi(s, t)$



HL: Query answering

Distance query answering:

- Store distance $c(\pi(v, w))$ for each label $w \in L(v)$
- $\min_{v \in L(s) \cap L(t)} (c(\pi(s, v)) + c(\pi(v, t)))$

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Space consumption: $\mathcal{O}(\sum_{v \in V} |L(v)|)$

Hence, the maximum individual label set size as well as the total labeling size should be **as small as possible**.